

R Programs: S3, S4 and Reference Classes

S3 Class - Banking System

```
# S3 Banking System

bank<-
function(acc,name,balance){structure(list(acc=acc,name=name,balance=balance),class="Bank")}

deposit<-function(x,amt){x$balance<-x$balance+amt;x}
withdraw<-function(x,amt){if(x$balance>=amt)x$balance<-x$balance-amt;x}
print.Bank<-function(x)cat(x$acc,x$name,x$balance,"\n")

b<-bank(101,"Arun",5000)
b<-deposit(b,1000)
b<-withdraw(b,2000)
print(b)

> # S3 Banking System
> bank<-function(acc,name,balance){structure(list(acc=acc,name=name,balance=balance),class="Bank")}
> deposit<-function(x,amt){x$balance<-x$balance+amt;x}
> withdraw<-function(x,amt){if(x$balance>=amt)x$balance<-x$balance-amt;x}
> print.Bank<-function(x)cat(x$acc,x$name,x$balance,"\n")
> b<-bank(101,"Arun",5000)
> b<-deposit(b,1000)
> b<-withdraw(b,2000)
> print(b)
101 Arun 4000
```

S4 Class - Payroll System

```
setClass("Employee",slots=list(id="numeric",name="character",basic="numeric"))
setGeneric("gross",function(object)standardGeneric("gross"))
setMethod("gross","Employee",function(object)object@basic+0.2*object@basic+0.1*object@basic)

e<-new("Employee",id=1,name="Anu",basic=30000)
show(e)
```

gross(e)

```
> setClass("Employee", slots=list(id="numeric", name="character", basic="numeric"))
> setGeneric("gross", function(object) standardGeneric("gross"))
[1] "gross"
> setMethod("gross", "Employee", function(object) object@basic+0.2*object@basic+0.1*object@basic)
> e<-new("Employee", id=1, name="Anu", basic=30000)
> show(e)
An object of class "Employee"
Slot "id":
[1] 1

Slot "name":
[1] "Anu"

Slot "basic":
[1] 30000

> gross(e)
[1] 39000
```

Reference Class - Banking System

```
Bank<-setRefClass("Bank",
fields=list(acc="numeric", name="character", balance="numeric"),
methods=list(
deposit=function(a){balance<<-balance+a},
withdraw=function(a){if(balance>=a)balance<<-balance-a},
display=function(){cat(acc,name,balance,"\n")}
))
b<-Bank$new(acc=101,name="Kumar",balance=10000)
b$deposit(5000);b$withdraw(3000);b$display()
```

```
> Bank<-setRefClass("Bank",
+               fields=list(acc="numeric", name="character", balance="numeric"),
+               methods=list(
+                 deposit=function(a){balance<<-balance+a},
+                 withdraw=function(a){if(balance>=a)balance<<-balance-a},
+                 display=function(){cat(acc,name,balance,"\n")}
+               ))
> b<-Bank$new(acc=101,name="Kumar",balance=10000)
> b$deposit(5000);b$withdraw(3000);b$display()
101 Kumar 12000
```

Reference Class - Payroll System

```
Payroll<-setRefClass("Payroll",
fields=list(id="numeric", name="character", basic="numeric"),
methods=list(
gross=function(){basic+0.2*basic+0.1*basic},
display=function(){cat(id,name,gross(),"\n")}
))
```

```
p<-Payroll$new(id=1,name="Meena",basic=40000)
```

```
p$display()
```

```
> Payroll<-setRefClass("Payroll",  
+                       fields=list(id="numeric",name="character",basic="numeric"),  
+                       methods=list(  
+                         gross=function(){basic+0.2*basic+0.1*basic},  
+                         display=function(){cat(id,name,gross(),"\n")}  
+                       ))  
> p<-Payroll$new(id=1,name="Meena",basic=40000)  
> p$display()  
1 Meena 52000
```